

Caleb Bellsmith

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TECHNICAL SKILLS

Languages: C++, HTML, L^AT_EX, Visual Basic

Software/Tools: SolidWorks, Fusion360, AutoCAD, VSCode, CLion, Bamboo, Arduino IDE

Fabrication: CNC Machining, 3D Printing, Soldering, Manual Prototyping

PROJECTS

Mechanical Exoskeleton

SolidWorks, Prototyping

Personal Project

2025 – Present

- Created a mechanical orthotic exoskeleton aimed at rehabilitating hand and wrist injuries.
- Optimized design for minimal weight and tool-less adjustability using SolidWorks modeling.

EMG-Activated Hand Exoskeleton

ESP32, C++, EMG Sensors

Independent Project

Dec 2025 – Present

- Developing a motorized orthotic using EMG sensors to detect neuromuscular signals for automated hand control.

Custom Cedar Strip Canoe

CAD, CNC Machining

Personal Project

2024 – Present

- Computer-modeled and fabricated a custom cedar strip canoe from scratch.
- Integrated CNC-machined internal molds with traditional woodworking to achieve high structural efficiency.

Adjustable Finger Prosthetic

User Testing, CAD, Iterative Design

Independent Project

2025 – Present

- Developed a functional finger prosthetic for a patient with scleroderma.
- Conducted repeated user-testing sessions to iterate on comfort and durability with interchangeable tips.

EXPERIENCE

True North Bio-competition

Electrical Designer

2025 – Present

- Creating the electrical control framework for a lower-limb exoskeleton, utilizing EMG sensors to facilitate intuitive, signal-based movement for assistive applications

Biomechatronics Design Team

Mechanical Designer

2025 – Present

- Developing mechanical components for assistive orthotic devices, focusing on structural integrity and wearability.

UWaterloo Space Soldering Team

Mechanical Designer

2025 – Present

- Designing mechanical housings and specialized hardware for microgravity soldering mechanisms.

BioTEC Conference

Sponsorship Lead

Dec 2025 – Present

- Working on corporate outreach and securing partnerships for Canada's largest student run biotechnology and entrepreneurship conference.

EDUCATION EXPERIENCE

University of Waterloo

Waterloo, ON

B.A.Sc. in Biomedical Engineering

Class of 2030

Honors: Governor General's Academic Medal, Sir Isaac Newton Physics Award, TDCH Valedictorian, HHCS Valedictorian.

WEEF Representative

Waterloo Engineering Endowment Foundation

Sept 2025 – Present

- Advocated for student project funding by evaluating technical proposals and representing class interests to the faculty.

Personal Tutor

Mathematics and Physics

Sept 2023 – June 2025

- Provided high school tutoring for students in calculus, physics, biology, and chemistry.